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Steps to Creating Unit Test Environment:

sudo env=containersLocal docker compose -f docker-compose.dev.yml up -d --build

Be sure that you are in the root directory(Bucstop_WebApp, Team-3-BucStop_APIGateway, etc) of where the .sln file lives (BucStop.sln,APIGateway.sln,...).

These commands are the commands I used to create the unit testing environment for the Bucstop_WebApp. For other root directories you will have to change the inputs to the commands, but the patterns should be the same.

Step	Action	Command (from Solution Root)	Purpose
1. Create Test Directory	Create the dedicated folder for tests.	mkdir Bucstop_WebApp.UnitTests	Keeps test code separate from application code.
2. Navigate into Test Directory	Change the current directory.	cd Bucstop_WebApp.UnitTests	Allows subsequent commands to target the new folder.
3. Create xUnit Project	Generate the xUnit project structure.	dotnet new xunit	Uses the official .NET template to create the .csproj file with xUnit references.
4. Add CORRECT Project Reference	Link the test project to the main application.	dotnet add reference ../BucStop/BucStop.csproj	This crucial step allows the tests to see and use classes (Controllers, Services, Models) from the main BucStop project.
5. Navigate Back to Root	Return to the solution directory.	cd ..	Prepares for the next command, which requires access to the .sln file.
6. Add Project to Solution	Register the new project with the solution file.	dotnet sln add Bucstop_WebApp.UnitTests/Bucstop_WebApp.UnitTests.csproj	Integrates the test project into your solution, making it visible in Visual Studio and included in solution-wide builds.
7. Verify Setup	Execute all tests in the solution.	dotnet test	Confirms the project builds and the default test runs successfully.

After running dotnet test you should see a bunch of output and at the bottom of it, it will say,"Test summary: total: 1, failed: 0, succeeded: 1, skipped: 0, duration: 0.6s"

This test corresponds with the [UnitTest1.cs](#) file it automatically makes for you.